DARWIN AND EVOLUTION

Module map

- Module 12: Darwin and Evolution
- Connected lab: lab-12_darwin-evolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 12

Historical explanation for adaptation

Variation and heritability

Fitness as reproductive success

Natural selection mechanism

Evidence for common ancestry

lab-12_darwin-evolution.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

DARWIN AND EVOLUTION

Learning objectives

- Explain natural selection using variation, heritability, and reproductive success.
- Define fitness in evolutionary terms.
- Distinguish individual change from population evolution.
- Use multiple evidence types to support common ancestry.
- Apply natural selection to a concrete trait example.

OBJECTIVE 1
Explain natural selection using variation, heritability, and reproductive success.

OBJECTIVE 2
Define fitness in evolutionary terms.

OBJECTIVE 3
Distinguish individual change from population evolution.

OBJECTIVE 4
Use multiple evidence types to support common ancestry.

OBJECTIVE 5
Apply natural selection to a concrete trait example.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

DARWIN AND EVOLUTION

Topic sequence

- Historical explanation for adaptation: Darwinian evolution explains adaptation through descent with modification.
- Variation and heritability: Natural selection requires variation, heritability, and unequal reproductive success.
- Natural selection mechanism: Fitness means relative reproductive success in a particular environment.
- Evidence for common ancestry: Fossils, homology, biogeography, and molecular evidence support common ancestry.
- Fitness as reproductive success: Evolution changes populations over generations, not individual organisms during life.

TOPIC 1
Historical explanation for adaptation

TOPIC 2
Variation and heritability

TOPIC 3
Natural selection mechanism

TOPIC 4
Evidence for common ancestry

TOPIC 5
Fitness as reproductive success

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Evidence Map

- Module focus: Darwin and Evolution
- Central claim: Darwin and Evolution explains natural selection from variation, heritability, fitness, and ancestry evidence.
- Key nodes to connect: Historical explanation for adaptation, Variation and heritability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-12_darwin-evolution.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Reasoning Sequence

- Module focus: Darwin and Evolution
- Trace the model: Historical explanation for adaptation -> Variation and heritability -> Natural selection mechanism -> Evidence for common ancestry
- Inputs become outputs: Historical explanation for adaptation, Variation and heritability -> Explain natural selection using variation, heritability, and reproductive success., Define fitness in evolutionary terms.
- Feedback check: lab-12_darwin-evolution.md evidence checks whether students can explain natural selection mechanism

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

DARWIN AND EVOLUTION

Terms as evidence handles

- Evolution: Change in heritable traits of populations across generations.
- Natural selection: Nonrandom differences in survival or reproduction tied to heritable traits.
- Adaptation: Heritable trait shaped by selection in a context.
- Fitness: Relative reproductive success.
- Common ancestry: Shared origin of lineages through descent.
- Homology: Similarity due to shared ancestry.

EVOLUTION
Change in heritable traits of populations across generations.

NATURAL SELECTION
Nonrandom differences in survival or reproduction tied to heritable traits.

ADAPTATION
Heritable trait shaped by selection in a context.

FITNESS
Relative reproductive success.

COMMON ANCESTRY
Shared origin of lineages through descent.

HOMOLOGY
Similarity due to shared ancestry.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

DARWIN AND EVOLUTION

Lab connection

- Lab file: lab-12_darwin-evolution.md
- Lab evidence should test or illustrate: Natural selection mechanism
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Historical explanation for adaptation

EVIDENCE
lab-12_darwin-evolution.md

CLAIM
Explain natural selection using variation, heritability, and reproductive success.

Teaching note: Make the lab connection explicit before students encounter the activity.

DARWIN AND EVOLUTION

Contrast check

- Do not stop at: Darwinian evolution explains adaptation through descent with modification.
- Stronger explanation: Evolution changes populations over generations, not individual organisms during life.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Darwinian evolution explains adaptation through descent with modification.

DEEPER
Evolution changes populations over generations, not individual organisms during life.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Retrieval and Lab Check

- Module focus: Darwin and Evolution
- Retrieval prompt: What are the required ingredients for natural selection?
- Answer check: Explain natural selection using variation, heritability, and reproductive success.
- Required terms: Evolution, Natural selection, Adaptation
- Lab connection: lab-12_darwin-evolution.md; lab-12_darwin-evolution.md supplies evidence for Darwinian evolution evidence from trait and environment comparisons.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

DARWIN AND EVOLUTION

Practice quiz bridge

- 1. Which statement best defines Evolution in Module 12?
- 2. Which learning goal best supports the topic "Variation and heritability"?
- 3. How should lab-12_darwin-evolution.md support Module 12?
- 4. Which retrieval move best prepares a student for Darwin and Evolution?

QUIZ 1
A

QUIZ 2
B

QUIZ 3
C

QUIZ 4
D

Teaching note: Use the quiz as formative feedback. Answer key: 1=A, 2=B, 3=C, 4=D

DARWIN AND EVOLUTION

Synthesis and exit ticket

- Exit claim: explain historical explanation for adaptation using evidence from lab-12_darwin-evolution.md.
- Must include: Evolution, Natural selection, Adaptation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Historical explanation for adaptation

EVIDENCE
lab-12_darwin-evolution.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.